

Trustpilot Interest Extraction

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1. Executive Summary

This document outlines a Proof of Concept (POC) for extracting actionable user interests from unstructured Trustpilot reviews. By moving beyond rigid keyword extraction and adopting an Agentic GraphRAG architecture, the system will autonomously extract relational data, group it into thematic communities, and allow AI agents to navigate the resulting knowledge graph to answer complex business queries.

2. Data Schema & Ingestion

The system will ingest the provided 2021 review dataset utilizing the following schema structure to establish the foundational nodes and edges of the graph:

- **Review_id:** Unique identifier (serves as the Primary Key for source nodes).
- **Created_date:** Used for temporal edge weighting and time-series filtering.
- **Star rating:** Provides sentiment validation across defined communities.
- **Review title & Review text:** The primary unstructured text used for entity extraction.
- **Review source:** Metadata for segmenting organic vs. invited reviews.
- **Language:** Enables multilingual routing and localization clustering.

3. Pipeline Architecture

Phase 1: Entity & Triplet Extraction (Nodes & Edges)

Instead of matching flat keywords, a Large Language Model (LLM) will process the review text to extract **Aspect-Opinion-Sentiment** triplets. This allows the system to capture the precise context of a user's statement.

Extraction Example:

A review stating *"The checkout page crashes on mobile"* is processed and extracted as a graph relationship:

(Checkout Page) - [FAILS_ON] -> (Mobile App)

Phase 2: Community Detection & Interest Definition

The extracted triplets form a massive Knowledge Graph. We define macro and micro "Interests" not by single keywords, but by semantic clusters of related concepts.

- **Clustering Algorithm:** Leiden or Louvain community detection groups highly interconnected nodes into distinct clusters (e.g., grouping "Cart," "Credit Card," and "Apple Pay").
- **Interest Summarization:** An LLM reads the entities within a cluster and generates a human-readable

Phase 3: Query Routing & Agentic Execution

Users will query the system in natural language. A routing LLM acts as the orchestrator, determining the optimal graph execution path based on the user's intent:

- **Global Search (Map-Reduce):** Deployed for thematic, open-ended questions (e.g., *"What are the top 3 complaints from 1-star reviews?"*). The system aggregates and synthesizes the pre-computed Community Summaries.
- **Local Search (Text-to-Cypher):** Deployed for specific root-cause drill-downs (e.g., *"Why is Apple Pay failing?"*). The LLM translates the query into a strict graph language to retrieve the exact 1-hop and 2-hop neighborhood around the specific entity node.

4. Example Agentic Workflow

An autonomous agent, utilizing a ReAct (Reason + Act) framework, will handle complex investigative tasks without human intervention. The table below illustrates a standard trace execution:

Phase	Action / Input	System Process & Observation
1. Prompt	User requests: <i>"Identify the most damaging emerging interest from 2021."</i>	The Agent initializes the reasoning loop and analyzes available graph tools.
2. Global Query	Agent calls <code>Query_Global_Communities</code> filtered by 1 and 2-star ratings.	The Graph returns Cluster 12. The summarized interest is identified as: <i>"Unexpected auto-renewal charges."</i>
3. Drill-down	Agent calls <code>Query_Local_Entity</code> on the "auto-renewal" node to find root causes.	The Graph retrieves connected edges: <code>(auto-renewal) <- [CAUSED_BY] - (hidden toggle switch)</code> .
4. Synthesis	Agent drafts the final explanatory response.	Combines the macro-trend (auto-renewal complaints) with the specific UI root cause (hidden toggles) and delivers the final text to the user.

Next Steps for Implementation: Configure the LLM prompts for reliable triplet extraction, establish the vector database/graph store (e.g., Neo4j), and define the LangChain toolsets for the routing agent.